

Name: Lau Yee Wen (A23CS0099) Section: 12

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

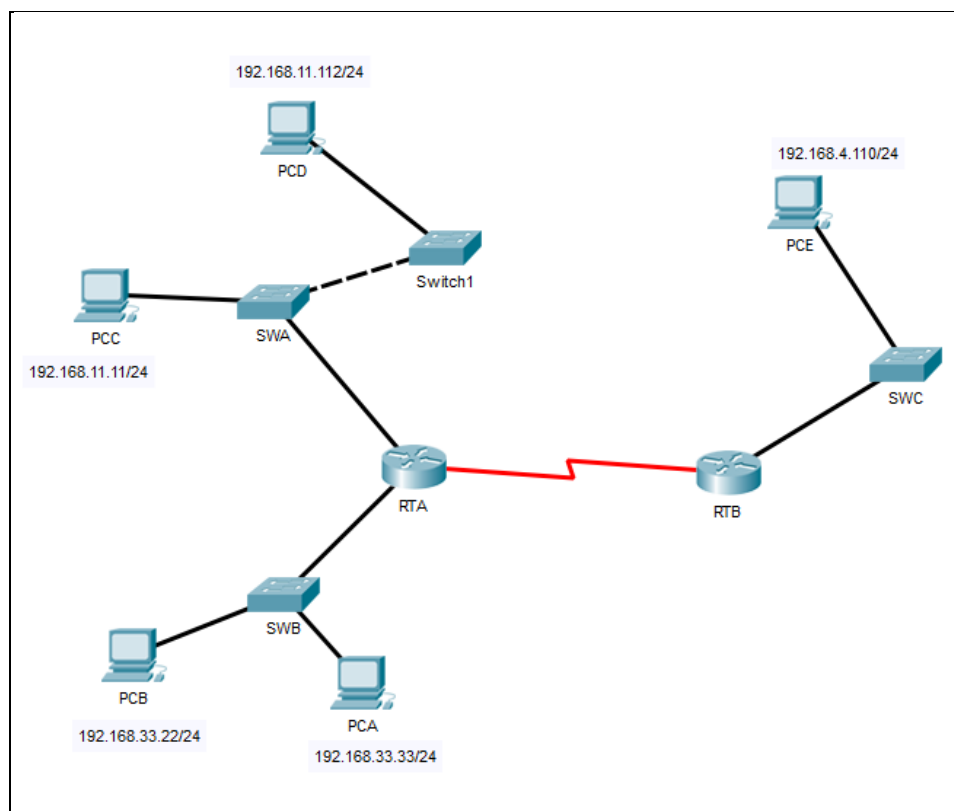


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.11.1 - 0002.4A00.0E91 ARPA FastEthernet1/0
Internet 192.168.33.1 - 000C.CF0C.593A ARPA FastEthernet0/0
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

SWA

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1     0002.4a00.0e91    DYNAMIC   Fa0/1
1     000c.8546.7d85    DYNAMIC   Fa1/1
```

SWAB

```
SWB>enable
```

```
SWB#show arp
```

```
SWB#show mac-address-table  
Mac Address Table
```

Vlan	Mac Address	Type	Ports
----	-----	-----	-----
1	000c.cf0c.593a	DYNAMIC	Fa0/1

SWC

```
SWC>enable
```

```
SWC#show arp
```

```
SWC#show mac-address-table  
Mac Address Table
```

Vlan	Mac Address	Type	Ports
----	-----	-----	-----
1	0001.977a.b614	DYNAMIC	Fa0/1

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type `arp -a` and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

PCA

```
Cisco Packet Tracer PC Command Line 1.0  
C:\>arp -a  
No ARP Entries Found  
C:\>
```

PCB

```
Cisco Packet Tracer PC Command Line 1.0  
C:\>arp -a  
No ARP Entries Found  
C:\>
```

PCC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

- e. What are your thoughts on the results?

No entry for all PCs because all the PCs didn't receive any ARP request.

There will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- a. Click **PCA** and click the Desktop tab > Command Prompt.
- b. Enter the **ping 192.168.33.22** command. This may take a few seconds.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time=2ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

- c. In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746       dynamic
```

- d. In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06       dynamic
```

- e. In the Command prompt of PCC, PCD abd PCE, type **arp -a**. Paste the result of this command here.

PCC

```
C:\>arp -a
No ARP Entries Found
```

PCD

```
C:\>arp -a
No ARP Entries Found
```

PCE

```
C:\>arp -a
No ARP Entries Found
```

Step 2: Generate traffic between PCC to all other PC except PCA.

- a. Click **PCC** and click the Desktop tab > Command Prompt.
- b. Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
```

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```

C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a

```

Internet Address	Physical Address	Type
192.168.11.1	0002.4a00.0e91	dynamic
192.168.11.112	0001.6462.0278	dynamic

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```

C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=2ms TTL=126
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126
Reply from 192.168.4.110: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 6ms, Average = 3ms

C:\>arp -a

```

Internet Address	Physical Address	Type
192.168.11.1	0002.4a00.0e91	dynamic
192.168.11.112	0001.6462.0278	dynamic

- e. Discuss the results you got from all the commands on PCC.

The ARP entries are revealed in the ARP table only after PCC pings PCB, PCD, and PCE. No ARP entries are found before these ping activities. The ARP table displays entries once PCB, PCD, and PCE receive the ARP request from PCC and subsequently receive pings from PCB, PCD, and PCE.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable  
RTA#show arp
```

```
RTA>enable  
RTA#show arp  
Protocol  Address          Age (min)  Hardware Addr  Type   Interface  
Internet  192.168.11.1      -         0002.4A00.0E91  ARPA   FastEthernet1/0  
Internet  192.168.11.11     3         00D0.D39A.C0D9  ARPA   FastEthernet1/0  
Internet  192.168.33.1      -         000C.CF0C.593A  ARPA   FastEthernet0/0  
Internet  192.168.33.22     3         0060.47EA.A746  ARPA   FastEthernet0/0  
RTA#
```

- g. At Router RTB, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTB>enable  
RTB#show arp
```

```
RTB>enable  
RTB#show arp  
Protocol  Address          Age (min)  Hardware Addr  Type   Interface  
Internet  192.168.4.1       -         0001.977A.B614  ARPA   FastEthernet0/0  
Internet  192.168.4.110     2         0060.702D.7C08  ARPA   FastEthernet0/0  
RTB#
```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
SWA>enable  
SWA#show arp  
  
SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0001.6462.0278    DYNAMIC Fa1/1
      1    0002.4a00.0e91    DYNAMIC Fa0/1
      1    000c.8546.7d85    DYNAMIC Fa1/1
      1    00d0.d39a.c0d9    DYNAMIC Fa2/1
SWA#
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    000c.cf0c.593a    DYNAMIC Fa0/1
SWB#
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```

SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0001.977a.b614    DYNAMIC Fa0/1
SWC#

Switch>enable
Switch#show arp

Switch#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0005.5e95.187b    DYNAMIC Fa0/1
Switch#

```

- d. Do switches use arp table? (Y/N)

Yes

- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

Switches use an ARP table to store the IP addresses and corresponding MAC addresses of network devices. This table serves the purpose of identifying the destination MAC addresses for communication with network nodes.

- f. What information does the command `show mac-address-table` gives?

To view MAC address table entries on a switch, the term "DYNAMIC" specifically pertains to displaying MAC addresses that have been learned by the switch during its operation.

Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot shows a Cisco Packet Tracer interface with a network topology. A switch (S1) is connected to a wireless router (WRS2) and three PCs (PC1, PC2, PC3). The topology includes subinterfaces for S1 and VLANs for each device. The PT Activity window is open, showing the 'Configuring Wireless LAN Access' activity. The 'Addressing Table' is displayed, and the completion status is 100/100.

Subinterfaces

- G0/0.10 172.17.10.1/24
- G0/0.20 172.17.20.1/24
- G0/0.88 172.17.88.1/24

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

PT Activity: 00:12:52

Completion: 100/100

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

The screenshot shows a Cisco Packet Tracer interface with a network topology. A switch (S1) is connected to a wireless router (WRS2) and three PCs (PC1, PC2, PC3). The topology includes subinterfaces for S1 and VLANs for each device. The PT Activity window is open, showing the 'Configuring Wireless LAN Access' activity. The 'Addressing Table' is displayed, and the completion status is 100/100.

Subinterfaces

- G0/0.10 172.17.10.1/24
- G0/0.20 172.17.20.1/24
- G0/0.88 172.17.88.1/24

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1
PC2	NIC	172.17.20.22	255.255.255.0	172.17.20.1
PC3	NIC	DHCP Assigned	DHCP Assigned	DHCP Assigned
WRS2	NIC	172.17.88.25	255.255.255.0	172.17.88.1

PT Activity: 00:16:44

Completion: 100/100

PC3

PhysicalConfigDesktopProgrammingAttributes

Link InformationConnectProfiles

BackStatisticsSave to Profile

Wireless Network Status

Radio Band	20MHz	Network Type	Mixed B/G/N
Wireless Network Name	WRS_LAN	IP Address	172.17.40.101
Wireless Mode	Infrastructure	Subnet Mask	255.255.255.0
Wide Channel	N/A	Default Gateway	172.17.40.1
Standard Channel	1 - 2.412GHz	DNS1	0.0.0.0
Security	WPA2-Personal	MAC Address	0050.0F3D.9808
Authentication	Auto		

2.4GHz



Signal Strength

Link Quality

Adapter is Active

Wireless-N Notebook Adapter

Wireless Network Monitor v1.0

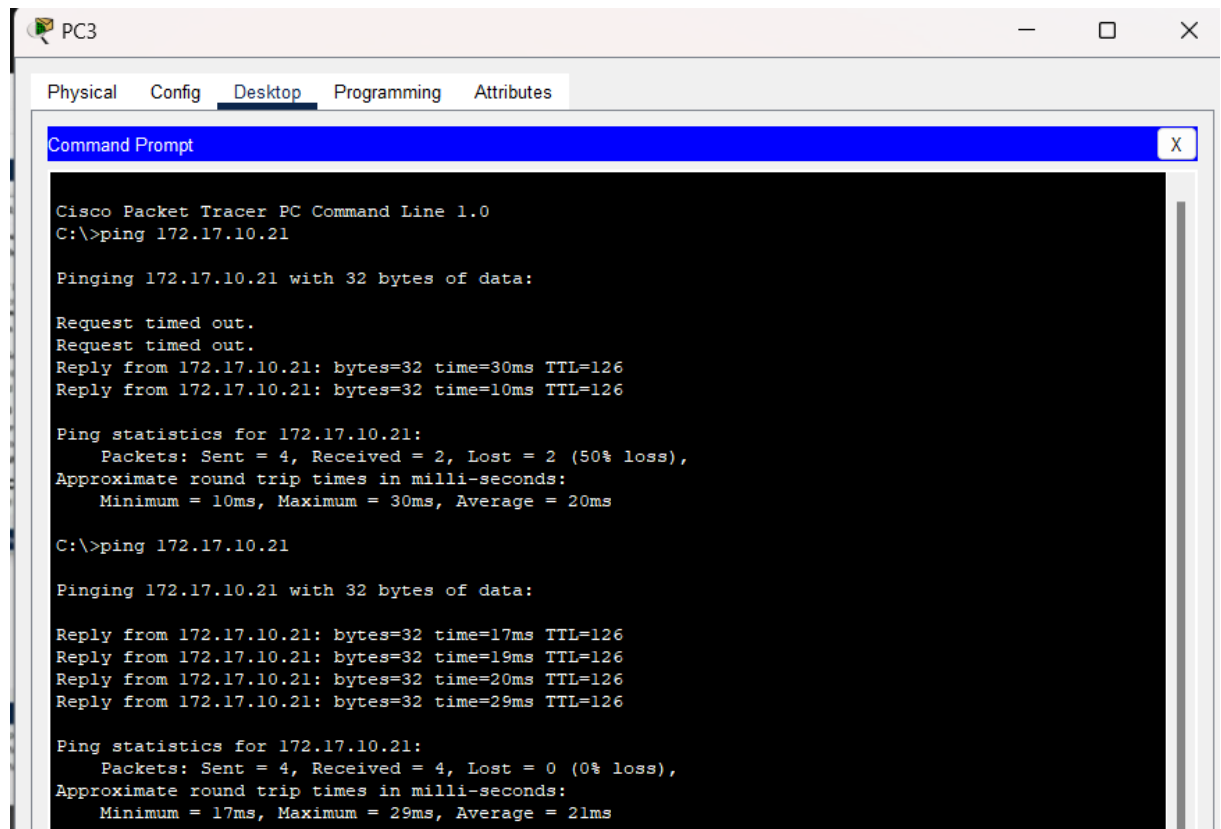
Model No. WPC300N

☐ Top

HKM_S11920

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Part 3: Verify Connectivity



The screenshot shows a PC3 window with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window titled "Command Prompt". The Command Prompt shows the output of a ping command to 172.17.10.21. The first attempt shows a 50% loss of packets, while the second attempt shows 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.17.10.21

Pinging 172.17.10.21 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 172.17.10.21: bytes=32 time=30ms TTL=126
Reply from 172.17.10.21: bytes=32 time=10ms TTL=126

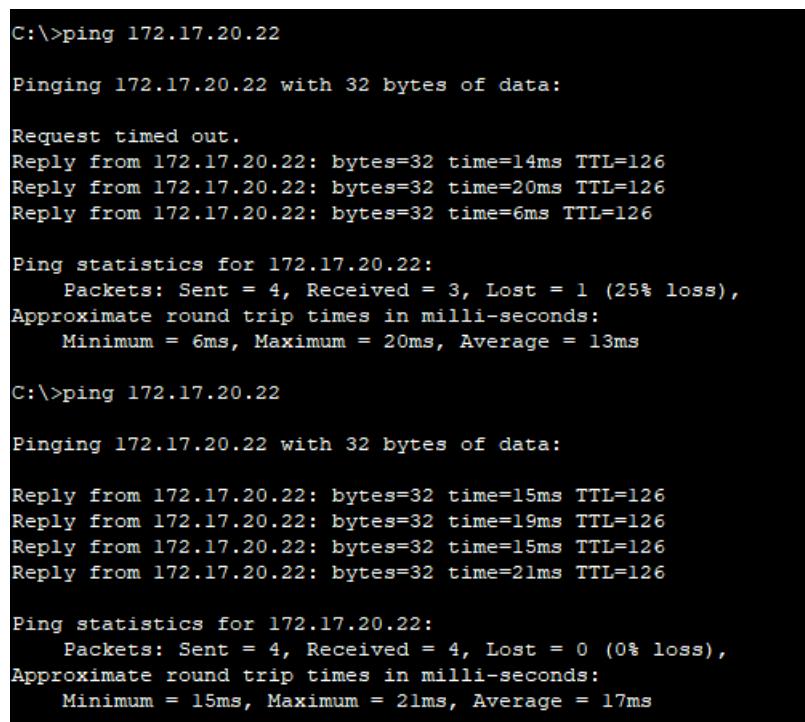
Ping statistics for 172.17.10.21:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 30ms, Average = 20ms

C:\>ping 172.17.10.21

Pinging 172.17.10.21 with 32 bytes of data:

Reply from 172.17.10.21: bytes=32 time=17ms TTL=126
Reply from 172.17.10.21: bytes=32 time=19ms TTL=126
Reply from 172.17.10.21: bytes=32 time=20ms TTL=126
Reply from 172.17.10.21: bytes=32 time=29ms TTL=126

Ping statistics for 172.17.10.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 29ms, Average = 21ms
```



The screenshot shows a Command Prompt window with the output of a ping command to 172.17.20.22. The first attempt shows a 25% loss of packets, while the second attempt shows 0% loss.

```
C:\>ping 172.17.20.22

Pinging 172.17.20.22 with 32 bytes of data:

Request timed out.
Reply from 172.17.20.22: bytes=32 time=14ms TTL=126
Reply from 172.17.20.22: bytes=32 time=20ms TTL=126
Reply from 172.17.20.22: bytes=32 time=6ms TTL=126

Ping statistics for 172.17.20.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 20ms, Average = 13ms

C:\>ping 172.17.20.22

Pinging 172.17.20.22 with 32 bytes of data:

Reply from 172.17.20.22: bytes=32 time=15ms TTL=126
Reply from 172.17.20.22: bytes=32 time=19ms TTL=126
Reply from 172.17.20.22: bytes=32 time=15ms TTL=126
Reply from 172.17.20.22: bytes=32 time=21ms TTL=126

Ping statistics for 172.17.20.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 21ms, Average = 17ms
```